Wave 6 — crescent [crescent_smoke_qd3] | Fitness 2.188 (SPG×A×(1+N)) | SPG 3.180 N/g | Est. weight 84.8 g

Detailed plate FEM (physics::fine DKT bending + T6 membrane) with per-element thickness field (taper, grinds, relief, lattice, fuller, flutes, texture). Capacities CERTIFIED = min over 0.25 / 0.125 mm² meshes; robust = min(20 °C at −0.1 mm stock, 100 °C yield)

AESTHETIC FLOOR: PASS
H 1.63 mm · EMD 0.005 · A 0.648

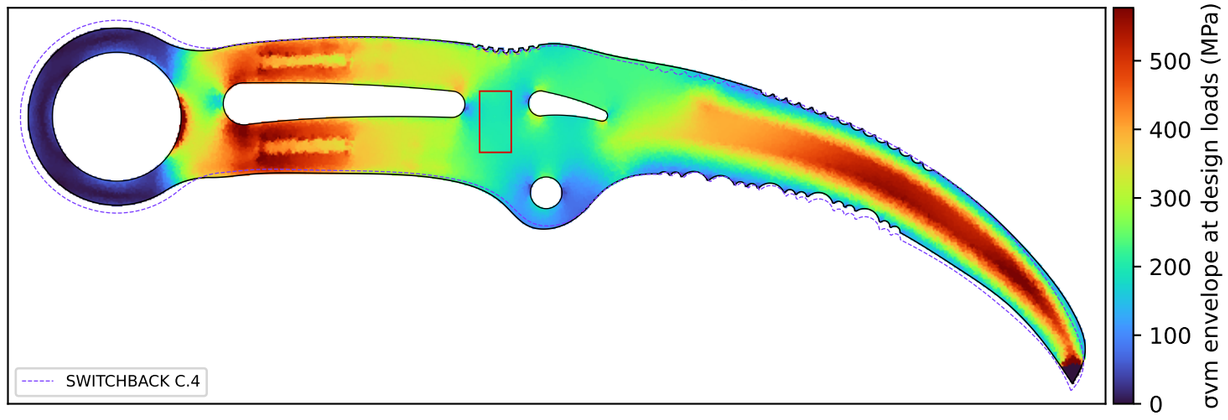
BLADE ≥ 3.375 in on all three: PASS
min 3.451 in

HARD GATES: PASS

FLOORS 180 / 260 / 1500 N (certified): PASS
lat 270 N · ring 14557 N

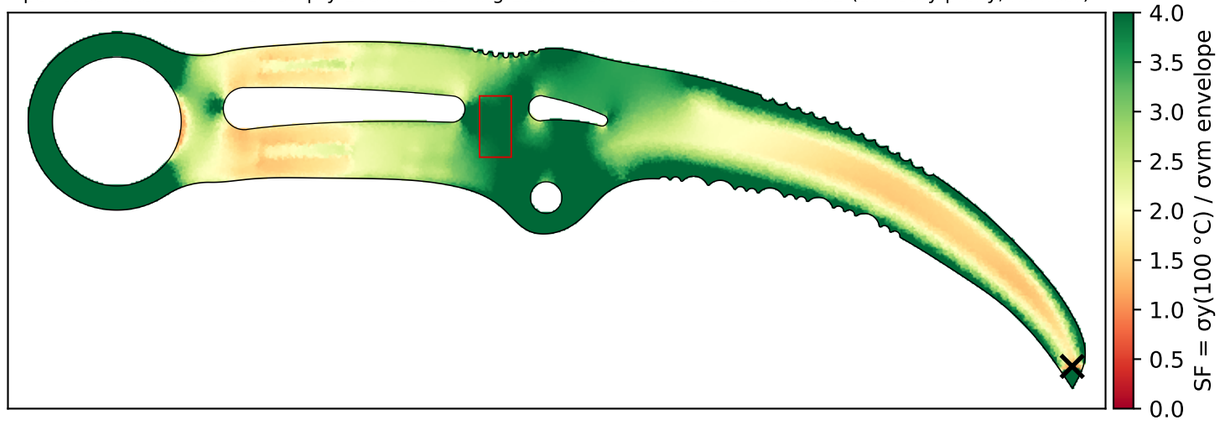
NOVELTY 0.061
(k-NN, aesthetic passers only)

von Mises envelope at design loads: lateral 260 N, pry 260 N, twist 1.26 N·m, ring pull 1500 N (4 mm tip-load zone excluded)

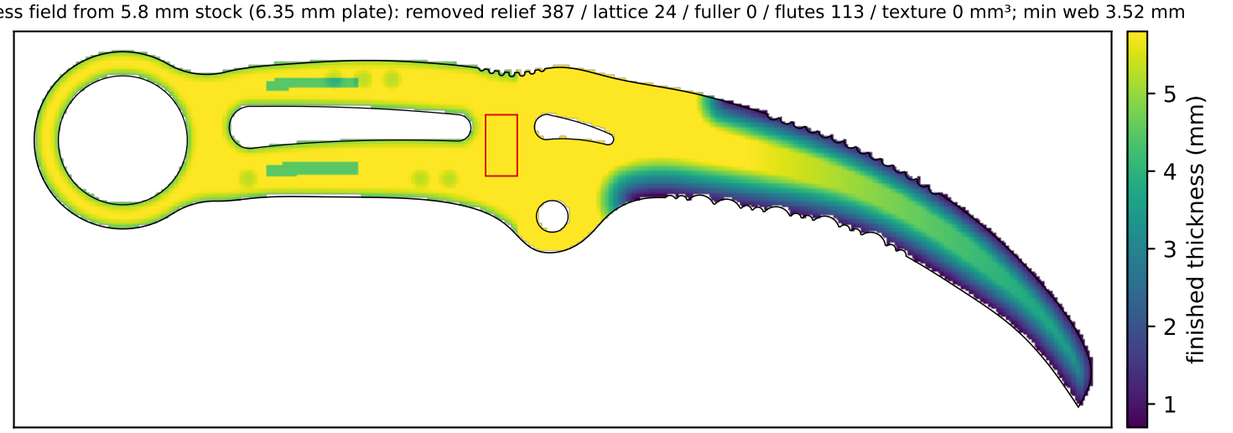


tip→heel	3.803 in (96.6)
horizontal reach	3.451 in (87.6)
edge arc inside / spine	3.987 / 3.958 in
grip / stock / Ti	61.4 mm / 5.8 mm / Ti-6Al-4V
weight (finished / blank)	84.8 g / 102.2 g
lateral 20 °C / robust	296 / 270 N
SF hiker 180 / hunter 260	1.50 / 1.04
ring pull 20 °C / robust (SF)	15962 / 14557 N (9.7)
tip pry 20 °C / twist 20 °C	698 N / 1.14 N·m
SF 100 °C lat / pry / ring / twist*	1.09 / 2.92 / 10.85 / 0.82
governing lateral at	(13, -3) mm
holder box (oriented)	198.9 × 98.8 mm @ -9.0°
aesthetic A / H / EMD	0.648 / 1.63 mm / 0.005
straight run / min blend R	17.5 / 13.3 mm
novelty / SPG	0.061 / 3.180 N/g

safety factor at 100 °C, envelope of all four cases (min 0.82 at x=188, y=-48 mm)
per case: lateral 260 N 1.09 · pry 260 N 2.92 · ring 1500 N 10.85 · twist 1.26 N·m 0.82 (advisory proxy, no floor)



thickness field from 5.8 mm stock (6.35 mm plate): removed relief 387 / lattice 24 / fuller 0 / flutes 113 / texture 0 mm³; min web 3.52 mm



novelty / QD map (all evaluations of the smoke run)

